



RESTAURANT SALES & CUSTOMER BEHAVIOR ANALYSIS

USING POWER BI

Prepared by:
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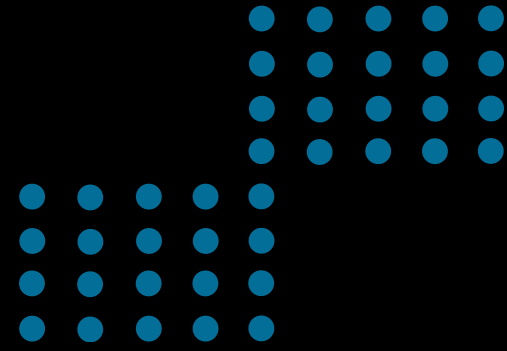
Project Objective:

To analyze restaurant sales performance, customer ordering behavior, and profitable combo meal opportunities using data visualization and market basket analysis.

Tools Used:

- Power BI
- Power Query Editor
- DAX Measures





Business Objectives

The following business questions were addressed during the analysis:

1. Identify the top-selling products and product categories.
2. Determine the days and hours with the highest sales activity.
3. Analyze customer ordering patterns and purchasing behavior.
4. Conduct Market Basket Analysis to identify profitable combo meal opportunities.
5. Generate business recommendations to improve sales and customer experience.



Methodology & Process of Analysis

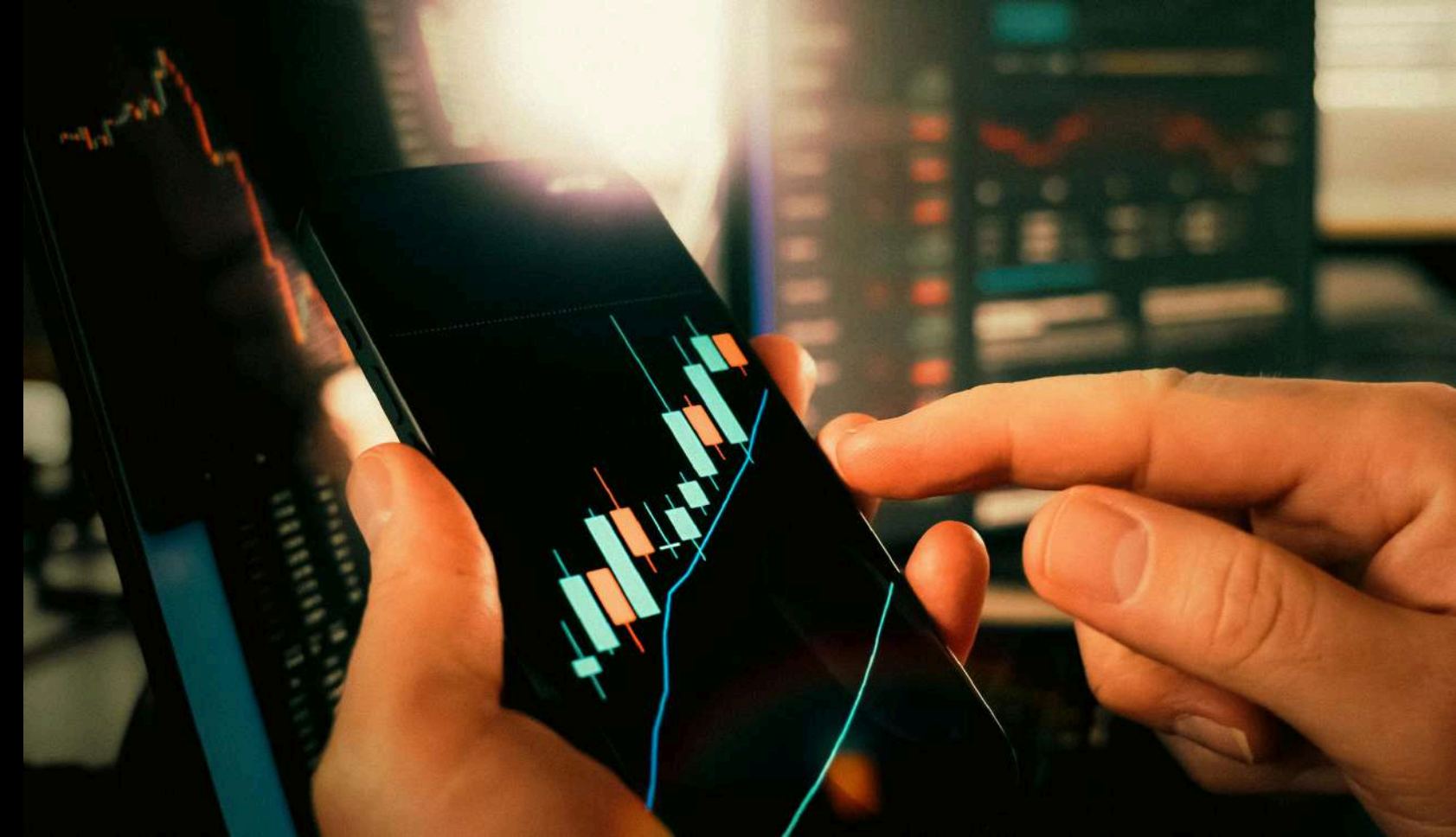
Data Preparation

- Cleaned and transformed raw restaurant sales data using Power Query Editor.
- Verified data consistency, missing values, and relationships between tables.
- Applied appropriate data types for date, sales, and quantity fields.

DAX Measures Created

The following measures were created to support analysis:

- Total Sales
- Total Discount
- Net Sales
- Average Sales
- Total Bills
- Total Items Sold
- Bills With Both
- Confidence
- Lift
- Qty Tie-In %



Analytical Techniques Used

- Sales Trend Analysis
- Product Performance Analysis
- Customer Order Pattern Analysis
- Market Basket Analysis
- Cross-Selling Opportunity Analysis

Identify the top-selling product or category of products.

TOP-SELLING PRODUCTS ANALYSIS

53K

Total Gross Sales

13K

Total Sales-Discout

39K

Total Net Sales

586

Total Dosas Sold

586

Dosas Sold

39K

Net Sales

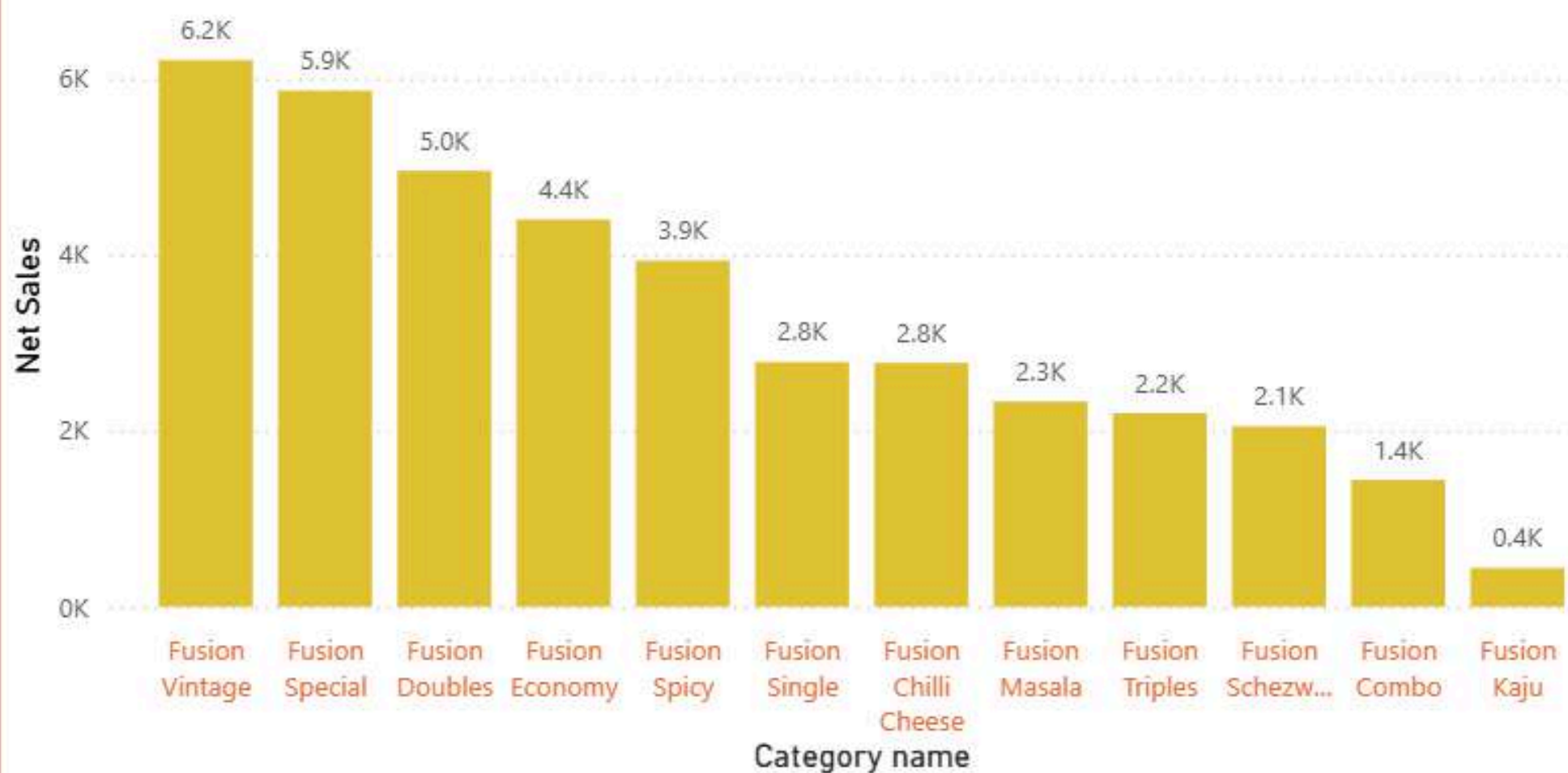
254

Total Bills

97.12

Average Sales

Net Sales by Category



Top Selling Dosa





Top-Selling Product Analysis

Top-Selling Products & Categories

KEY FINDINGS

- Pizza Dosa emerged as the highest revenue-generating product.
- Plain Dosa and Stump/Hatrick Dosa also showed strong sales performance (along with Mushroom Chilli Cheese, Paneer Chinese Mushroom, Paneer Cheese).
- Fusion Vintage and Fusion Special categories generated the highest category-level sales, followed by Fusion Doubles, and Fusion Economy.

INSIGHTS

- Customers showed strong preference toward fusion-style, paneer-based and cheese-based dosas.
- Premium fusion products generated higher revenue contribution than traditional items.
- Pizza Dosa acted as the strongest “anchor product” across the menu.

BUSINESS RECOMMENDATION

- Promote Pizza Dosa as a flagship menu item.
- Expand high-performing fusion categories with additional variants.

- Use best-selling products in combo meals and promotional offers.
- Consider reducing promotional focus on very low-performing products unless they serve a niche demand.

On which days are the sales higher?

OVERALL DAILY & HOURLY SALES ANALYSIS

53K

Total Gross Sales

13K

Total Sales-Discout

39K

Total Net Sales

586

Total Dosas Sold

254

Total Bills

97.12

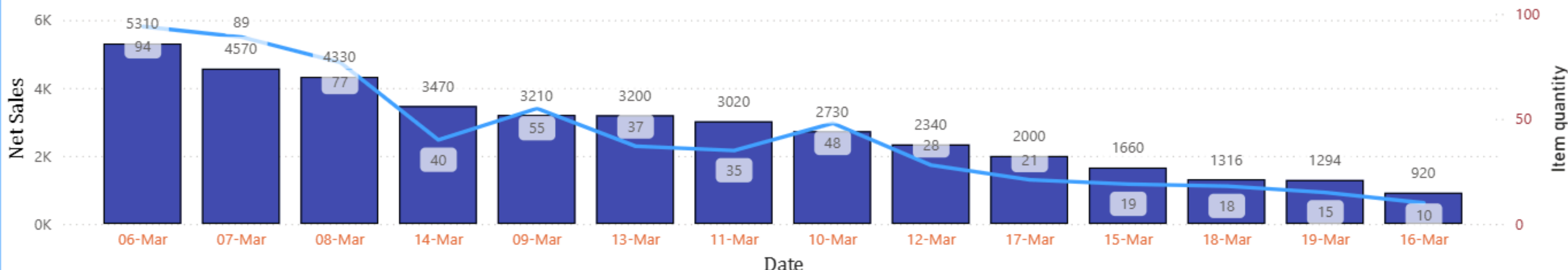
Average Sales

586

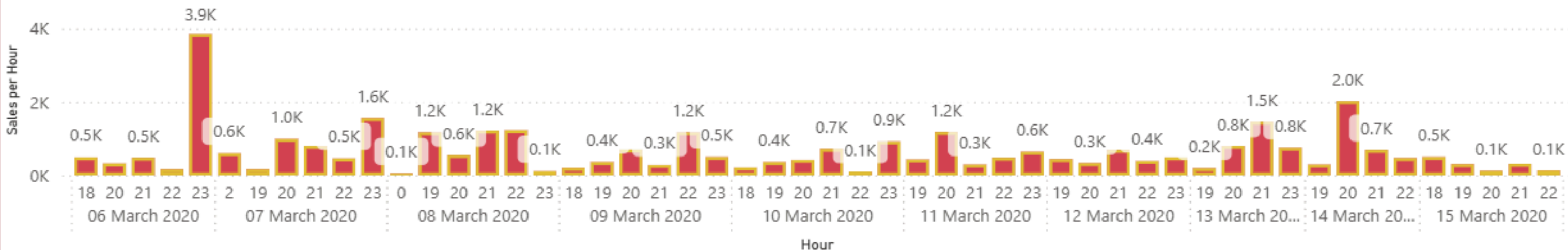
Dosas Sold

Date-wise Sales

Net Sales Item quantity



Hourly Sales



Date

06/03/2020

19/03/2020

39K

Net Sales



Daily & Hourly Sales Analysis

Overall Daily & Hourly Sales Analysis

Key Findings

- 6th March recorded the highest net sales and item quantity sold, followed by 7th and 8th March.
- Sales activity was consistently stronger between 8 PM and 11 PM.
- Late evening hours generated the highest average hourly sales.

Insights

- Customer demand is heavily concentrated during dinner hours.
- Weekend and early-month periods showed stronger restaurant activity.
- High evening traffic indicates strong potential for dinner combo promotions.



Business Recommendation

- Increase staffing during peak evening hours.
- Launch dinner-specific offers between 8 PM–11 PM & during slower sales periods to improve consistency.
- Use peak-hour sales trends for inventory planning and operational efficiency. Promote high-performing menu items during peak order windows.
- Investigate reasons behind declining sales trends after early March.
- Monitor repeat customer behavior and customer retention patterns.



Which products are selling more on particular days?

TOP-SELLING PRODUCTS PER DAY ANALYSIS pt.1

Date

06/03/2020

19/03/2020



39K

Net Sales

254

Total Bills

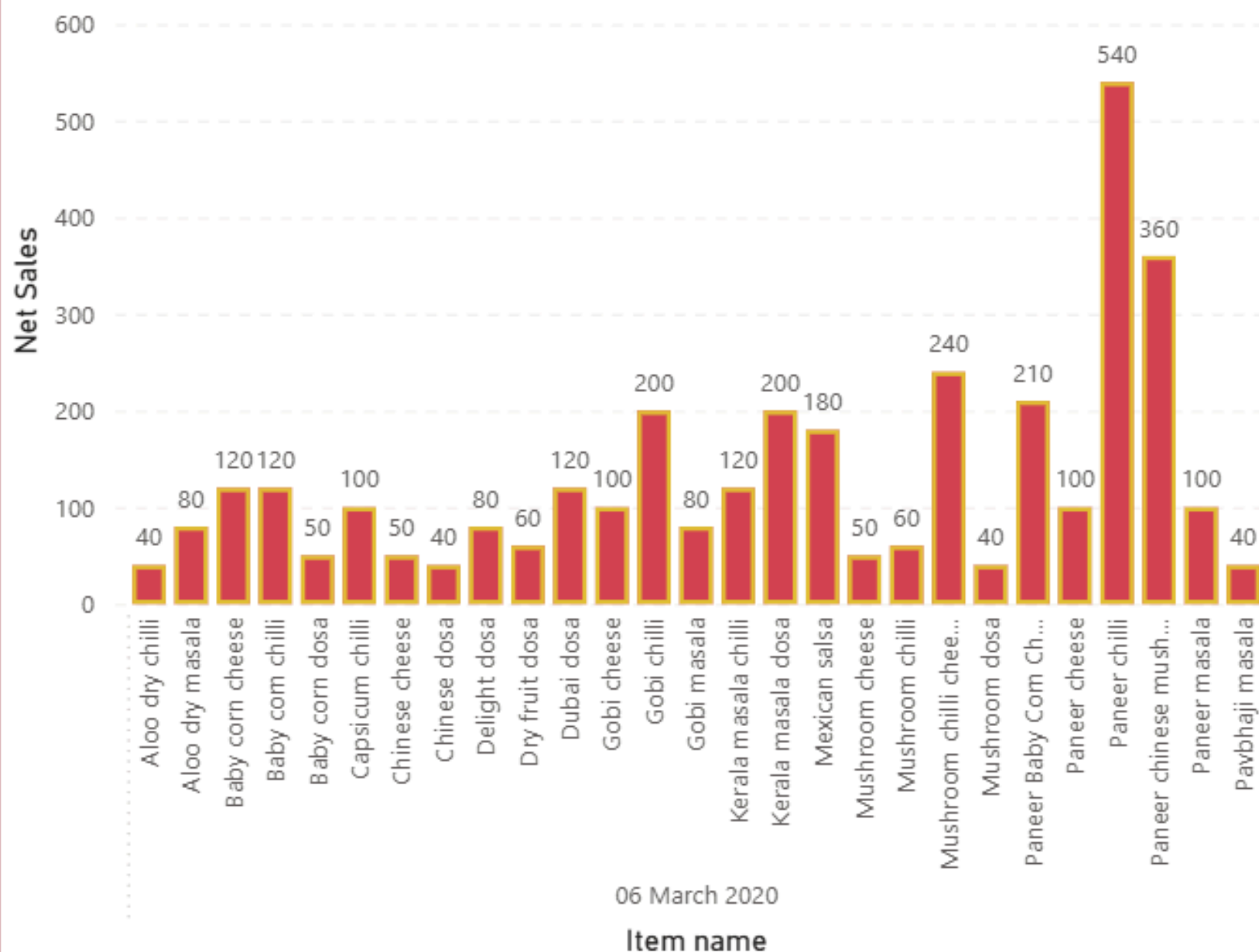
97.12

Average Sales

586

Dosas Sold

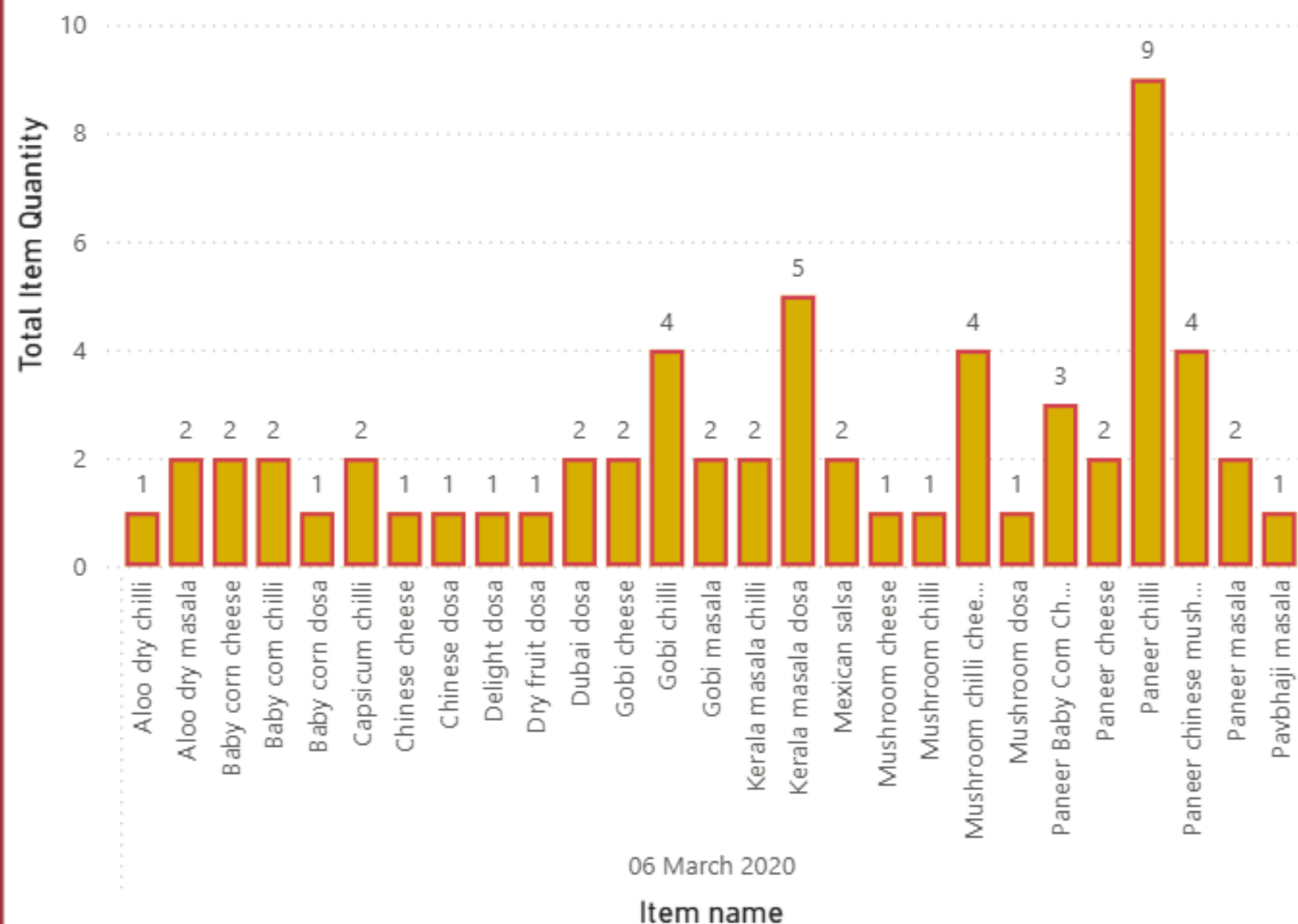
Top Selling Dosa



06 March 2020

Item name

Top Selling Dosa



06 March 2020

Item name



Top-Selling Products per Day



Part 1 - Product Performance by Day

KEY FINDINGS

- Pizza Dosa consistently remained one of the top-selling products across multiple days.
- Paneer-based and cheese-based products repeatedly appeared among daily best-sellers.
- Mushroom Chilli Cheese and Paneer Chinese Mushroom showed stable repeat demand.

INSIGHTS

- Customers consistently preferred spicy/schezwan, cheesy, and paneer-based combinations.
- Product demand remained relatively stable across multiple sales days.
- Fusion “comfort + indulgence” products demonstrated stronger repeat-purchase behavior than niche/experimental items.

COMBO STRATEGY

The strong pairing patterns suggest:

- Paneer + Cheese combos
- Schezwan + other dosa combos
- Mushroom + Cheese combos

BUSINESS RECOMMENDATION

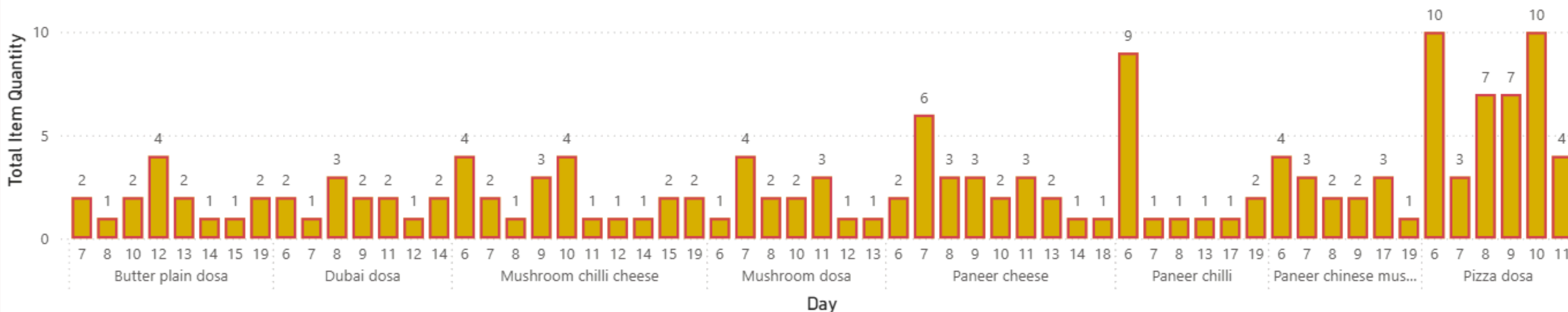
- Maintain strong stock availability for top-performing fusion (cheese, paneer, pizza dosa) products & Reduce waste by limiting rarely ordered items.
- Introduce rotating “Daily Bestseller” promotions.
- Highlight repeat-performing items on digital and physical menus.



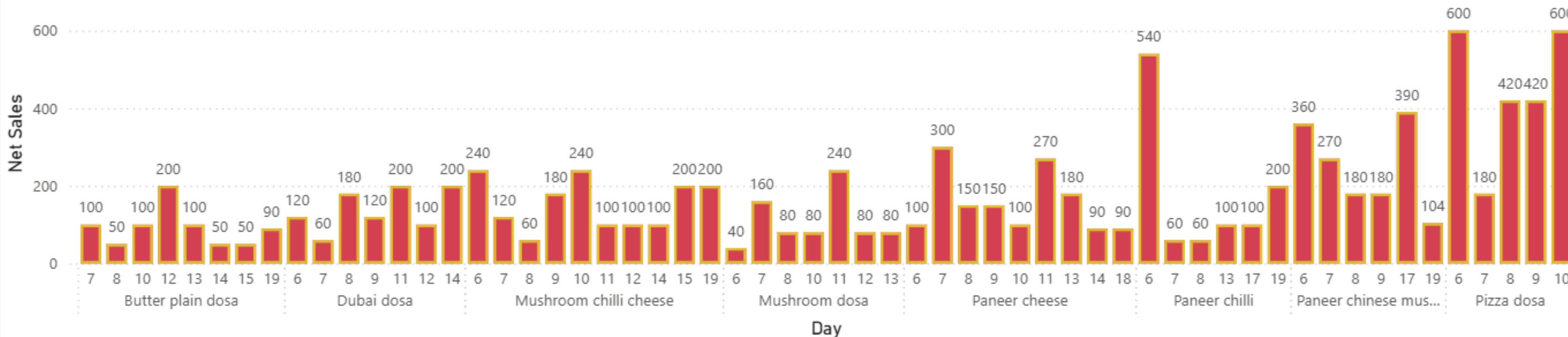
On which days are the top selling products selling more?

TOP-SELLING PRODUCTS PER DAY ANALYSIS pt.2

Top Selling Dosa



Top Selling Dosa





Top-Selling Products per Day

Part 2 - Product Sales Trends Across Days

KEY FINDINGS

This analysis highlights that top-selling products do not perform uniformly across all days, and instead show:

- strong demand spikes on specific dates
- recurring high-performance products across the timeline
- clear “peak vs low demand” cycles

- Pizza Dosa achieved the highest quantity sold across most dates.
- Plain Dosa displayed strong repeat ordering behavior despite being a basic menu item.
- Paneer Cheese and Mushroom Chilli Cheese showed strong recurring sales patterns.

A key pattern observed:

- High quantity days ≠ always highest net sales per product
- Some days show: high volume but lower per-item value; while others show fewer orders but higher-value items

INSIGHTS

- Basic menu items such as Plain Dosa act as strong supporting products.
- Customers frequently combine premium fusion products with simpler items.
- Consistent repeat purchases indicate strong customer satisfaction for top products.

Peak days are not driven by a single item — instead:

- multiple top products peak together
- indicating overall demand surges rather than isolated popularity

BUSINESS RECOMMENDATION

- Use Pizza Dosa and Plain Dosa as combo anchors. Always maintain buffer stock for these 2 along with Paneer Cheese variants.
- Promote high-repeat products through loyalty or recommendation systems.
- Create “Most Ordered Today” sections in menus.

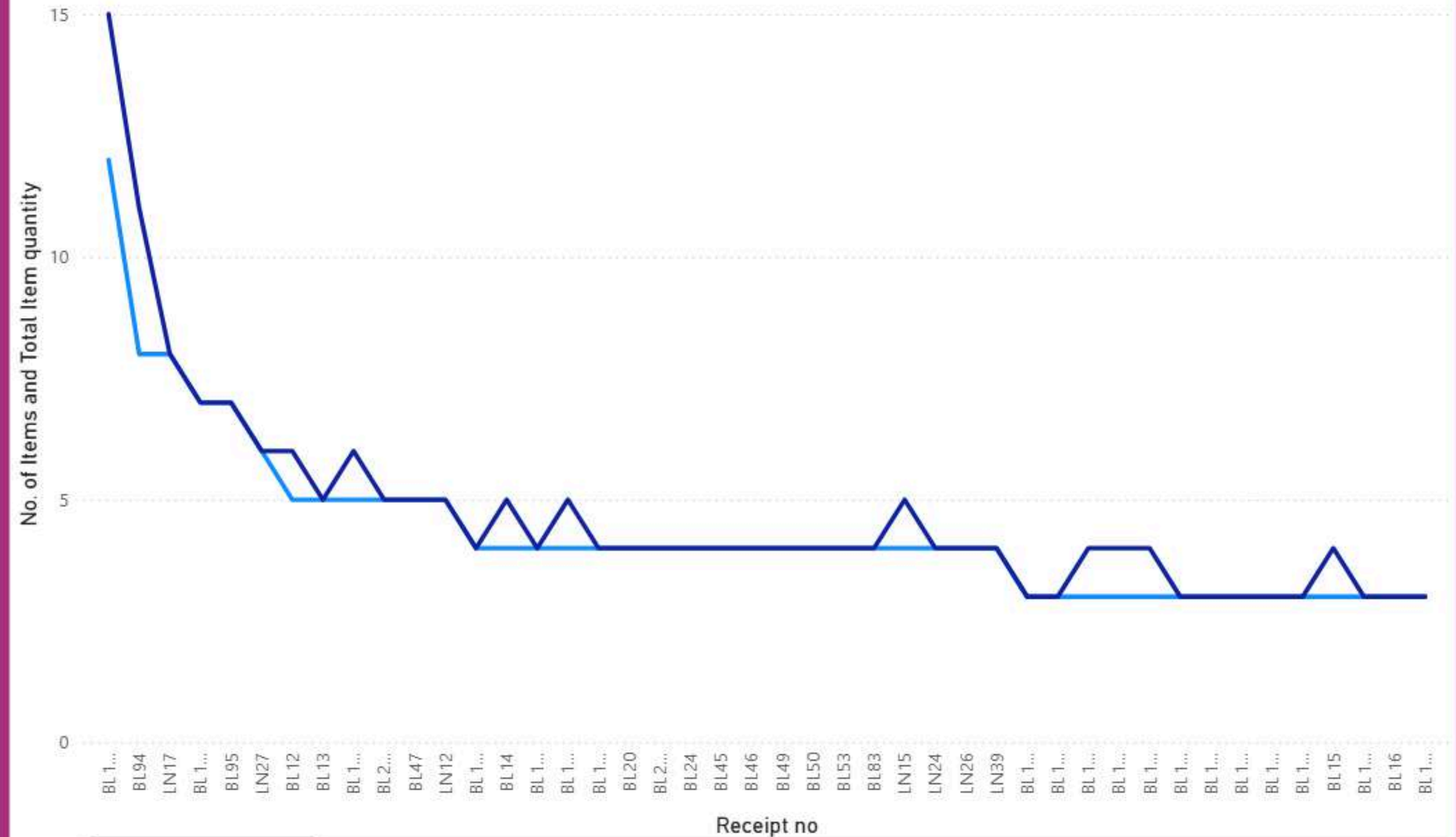


CUSTOMER ORDER PATTERN ANALYSIS pt. 1

Receipt no	No. of Items	Total Item quantity
BL175	12	15
BL94	8	11
LN17	8	8
BL110	7	7
BL95	7	7
LN27	6	6
BL12	5	6
BL13	5	5
BL174	5	6
BL213	5	5
BL47	5	5
LN12	5	5
BL106	4	4
BL14	4	5
BL158	4	4
BL184	4	5
BL189	4	4
BL20	4	4
BL209	4	4
BL24	4	4
BL45	4	4
BL46	4	4
BL49	4	4
BL50	4	4
BL53	4	4
BL83	4	4
LN15	4	5
LN24	4	4
LN26	4	4
LN39	4	4
BL107	3	3
Total	541	586

No. of Items & Total Item quantity per bill

● No. of Items ● Total Item quantity





Customer Order Pattern Analysis

Part 1 - Customer Order Quantity Analysis

KEY FINDINGS

- Most bills contained 1–3 items.
- Several high-value bills contained 5–12 items.
- Certain customers consistently ordered multiple products together.

INSIGHTS

- Customers prefer bundled or combination-style ordering rather than single-item purchases.
- Large orders indicate group dining or family-ordering behavior.
- High multi-item transactions create strong upselling opportunities.

BUSINESS RECOMMENDATION

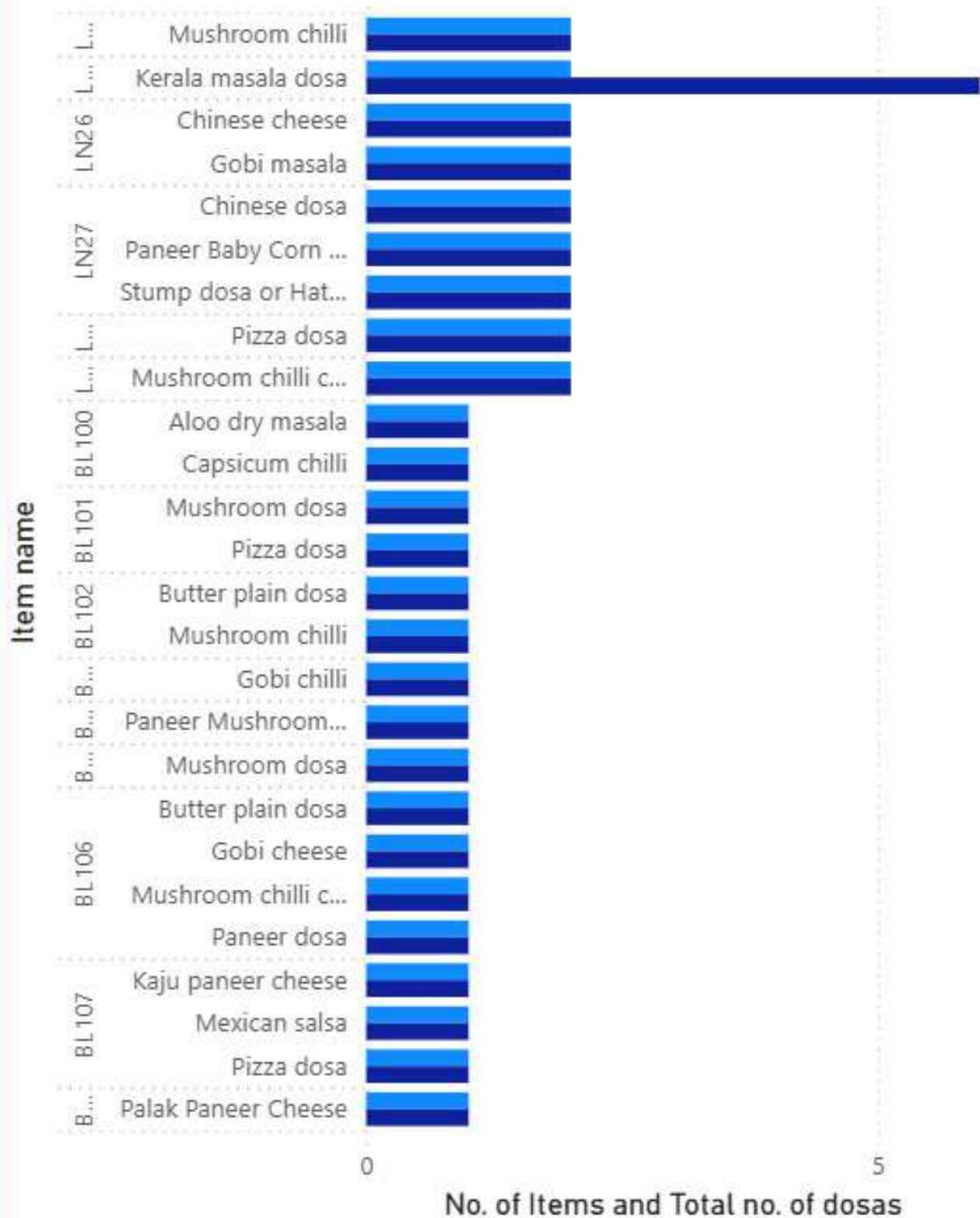
- Introduce family combo meals and sharing platters.
- Provide incentives for higher-order-value transactions.
- Recommend add-on products during checkout.



CUSTOMER ORDER PATTERN ANALYSIS pt. 2

Items Ordered Together

● No. of Items ● Total no. of dosas

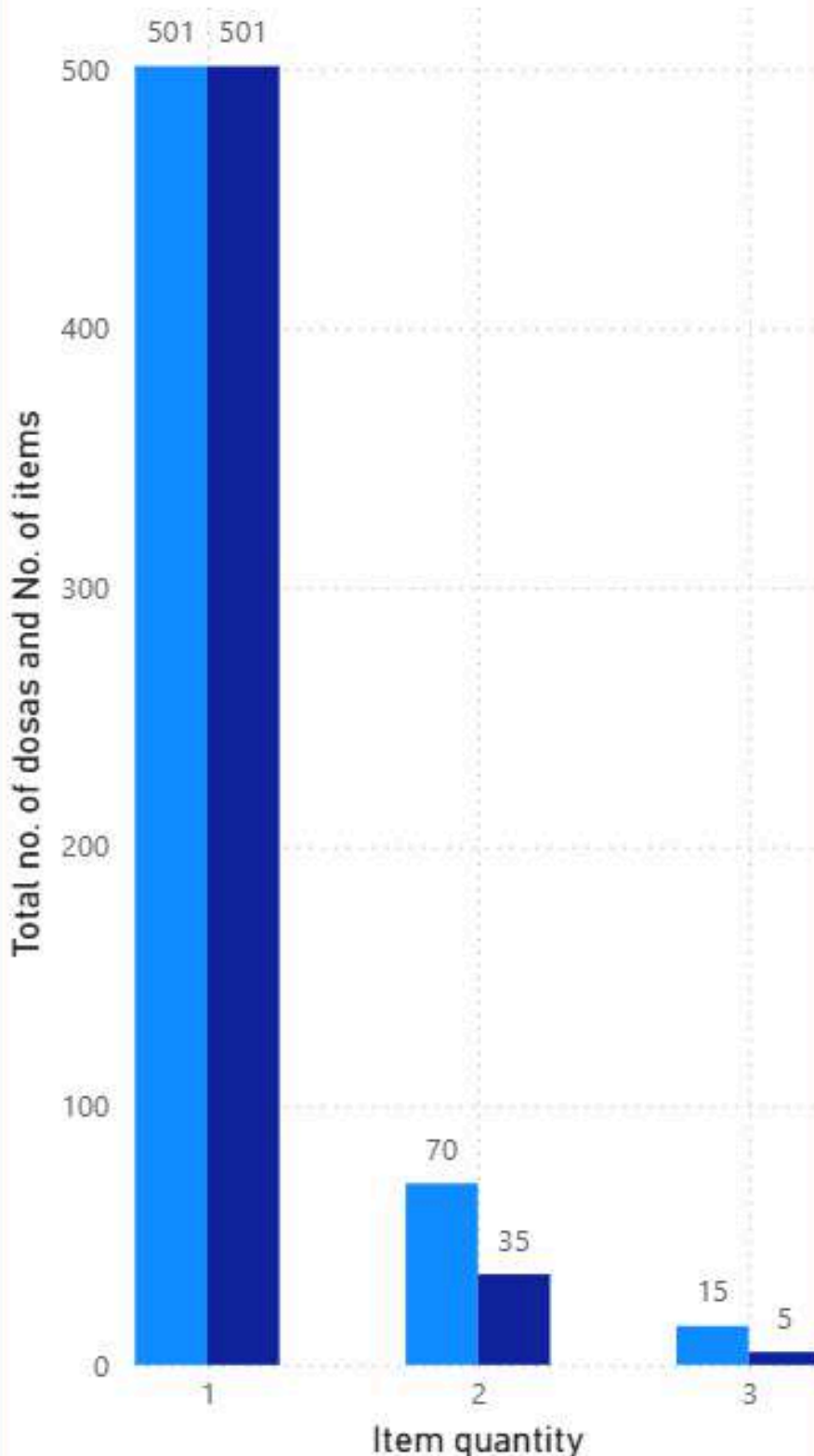


Receipt no No. of Items Total no. of dosas

BL175	12	15
American chopsyuey	1	2
Dry fruit dosa	1	1
Kaju paneer cheese	1	1
Kerala masala chilli	1	1
Mexican salsa	1	1
Mushroom chilli cheese	1	1
Paneer cheese	1	1
Pizza dosa	1	2
Plain dosa	1	2
Schezwan mixveg dosa	1	1
Schezwan spring roll dosa	1	1
Stump dosa or Hatrick dosa	1	1
BL94	8	11
Alleppey special	1	1
Chinese mushroom	1	2
Delight dosa	1	1
Mexican salsa	1	1
Mushroom chilli cheese	1	1
Paneer mushroom dosa	1	1
Pizza dosa	1	3
Schezwan paneer dosa	1	1
LN17	8	8
Baby corn chilli	1	1
Dubai dosa	1	1
Mushroom cheese	1	1
Paneer cheese	1	1
Paneer chilli	1	1
Pizza dosa	1	1
Plain dosa	1	1
Spring roll cheese	1	1
Total	541	586

Quantity Ordered

● Total no. of dosas ● No. of items





Customer Order Pattern Analysis

Part 2 - Items Ordered Together

KEY FINDINGS

Common item combinations included:

- Pizza Dosa + Paneer Cheese
- Mushroom Chilli Cheese + Paneer Cheese
- Pizza Dosa + Plain Dosa
- Paneer Chinese Mushroom + Paneer Cheese
- Schezwan Butter Onion + Plain Dosa

INSIGHTS

- Customers display strong preference for cheese-heavy and spicy combinations.
- Paneer-based items act as strong secondary purchases.
- Customers rarely build large bundles; instead they prefer: 1 base dosa + 1 fusion topping item
- Repeated pairing suggests strong cross-sell potential between core + fusion items

BUSINESS RECOMMENDATION

- Design combo meals around common item pairings.
- Use cross-selling recommendations within menus.
- Train staff to upsell frequently paired products.

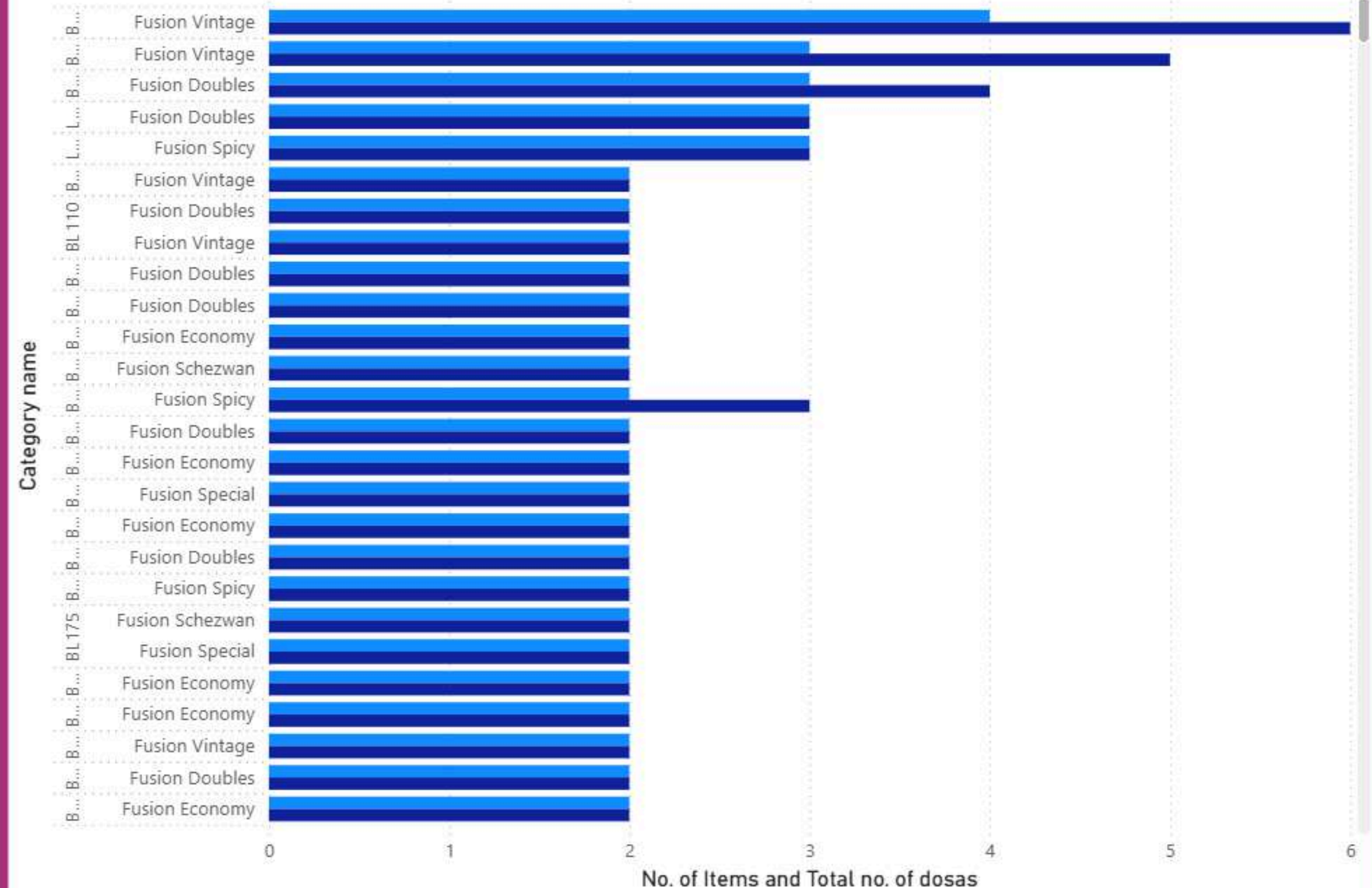


CUSTOMER ORDER PATTERN ANALYSIS pt. 3

Receipt no	No. of Items	Total no. of dosas
BL175	12	15
Fusion Vintage	4	6
Fusion Economy	1	2
Fusion Schezwan	2	2
Fusion Special	2	2
Fusion Chilli Cheese	1	1
Fusion Doubles	1	1
Fusion Spicy	1	1
BL94	8	11
Fusion Vintage	3	5
Fusion Combo	2	3
Fusion Chilli Cheese	1	1
Fusion Schezwan	1	1
Fusion Special	1	1
LN17	8	8
Fusion Doubles	3	3
Fusion Spicy	2	2
Fusion Economy	1	1
Fusion Special	1	1
Fusion Vintage	1	1
BL110	7	7
Fusion Doubles	2	2
Fusion Vintage	2	2
Fusion Masala	1	1
Fusion Schezwan	1	1
Fusion Special	1	1
BL95	7	7
Fusion Economy	2	2
Fusion Chilli Cheese	1	1
Fusion Masala	1	1
Fusion Special	1	1
Total	541	586

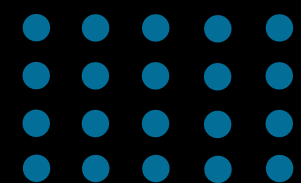
Items Ordered Together

● No. of Items ● Total no. of dosas





Customer Order Pattern Analysis



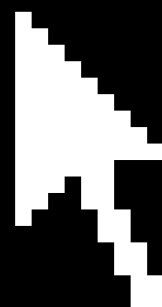
Part 3 - Category-Level Basket Patterns

KEY FINDINGS

- Fusion Vintage, Fusion Doubles, and Fusion Special categories frequently appeared together in customer baskets.
- Customers commonly combined premium fusion products with spicy and schezwan variants.

INSIGHTS

- Customers prefer variety-oriented ordering behavior.
- Fusion-style products dominate cross-category transactions.
- Premium fusion categories create strong cross-selling potential.



DOMINANT CATEGORY TRENDS

- Fusion Vintage → most frequently recurring across receipts
- Fusion Economy → consistently high-volume, budget-driven segment
- Fusion Spicy & Fusion Doubles → strong secondary contributors

BUSINESS RECOMMENDATION

- Create category-based combo offers.
- Introduce “Fusion Special” recommendation sections.
- Bundle high-performing categories into curated meal packages.



MARKET BASKET ANALYSIS

The Transaction Matrix (The Core Grid) -- Sales [Item Name] x Basket Product [Item Name] - Bills with common items from both tables & Confidence

Item name	Alleppey special		Aloo dry chilli		Aloo dry masala		Aloo dry masala cheese		Aloo Gobi		Aloo paneer		Aloo F
Item name	Bills With Both	Confidence	Bills With Both	Confidence	Bills With Both	Confidence	Bills With Both	Confidence	Bills With Both	Confidence	Bills With Both	Confidence	Bills W
Alleppey special	1	100.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	
Aloo dry chilli	0	0.00%	6	100.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	
Aloo dry masala	0	0.00%	0	0.00%	7	100.00%	0	0.00%	0	0.00%	0	0.00%	
Aloo dry masala cheese	0	0.00%	0	0.00%	0	0.00%	3	100.00%	0	0.00%	0	0.00%	
Aloo Gobi	0	0.00%	0	0.00%	0	0.00%	0	0.00%	4	100.00%	0	0.00%	
Aloo paneer	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	3	100.00%	
Aloo Plalak	0	0.00%	1	100.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	
American chopsuey	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	
American chopsuey cheese	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	
Baby corn cheese	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	1	100.00%	
Total	1	0.39%	6	2.36%	7	2.76%	3	1.18%	4	1.57%	3	1.18%	

The Volume Breakdown Table

Item name	Qty Sold With Basket Item	Qty Tie-In %
Alleppey special	1	1.00
Aloo dry chilli	6	1.00
Aloo dry masala	7	1.00
Aloo dry masala cheese	3	1.00
Aloo Gobi	4	1.00
Aloo paneer	3	1.00
Aloo Plalak	1	1.00
American chopsuey	3	1.00
American chopsuey cheese	7	1.00
Baby corn cheese	10	1.00
Baby corn chilli	4	1.00
Baby corn chilli cheese	12	1.00
Baby corn dosa	3	1.00
Total	586	1.00

The "Deep Dive" Cross-Sell Matrix - Sales [Item Name] x Basket Product [Item Name] - Lift

Item name	Alleppey special	Aloo dry chilli	Aloo dry masala	Aloo dry masala cheese	Aloo Gobi	Aloo paneer	Aloo
Alleppey special	254.00	0.00	0.00	0.00	0.00	0.00	
Aloo dry chilli	0.00	42.33	0.00	0.00	0.00	0.00	
Aloo dry masala	0.00	0.00	36.29	0.00	0.00	0.00	
Aloo dry masala cheese	0.00	0.00	0.00	84.67	0.00	0.00	
Aloo Gobi	0.00	0.00	0.00	0.00	63.50	0.00	
Aloo paneer	0.00	0.00	0.00	0.00	0.00	84.67	
Aloo Plalak	0.00	42.33	0.00	0.00	0.00	0.00	
American chopsuey	0.00	0.00	0.00	0.00	0.00	0.00	
American chopsuey cheese	0.00	0.00	0.00	0.00	0.00	0.00	
Baby corn cheese	0.00	0.00	0.00	0.00	0.00	8.47	
Baby corn chilli	0.00	0.00	0.00	0.00	0.00	0.00	
Baby corn chilli cheese	0.00	0.00	0.00	0.00	0.00	0.00	
Baby corn dosa	0.00	0.00	0.00	0.00	0.00	0.00	
Total	1.00	1.00	1.00	1.00	1.00	1.00	1.00



Market Basket Analysis

Market Basket Analysis & Combo Recommendations

KEY FINDINGS

- Pizza Dosa was the strongest basket anchor product.
- Plain Dosa, Paneer Cheese, and Mushroom Chilli Cheese displayed strong basket associations.
- Several item pairs showed extremely high Lift values, indicating strong product relationships.

STRONG SCALABLE COMBOS

- Pizza Dosa + Paneer Chilli
- Mushroom Chilli Cheese + Pizza Dosa
- Mexican Salsa + Pizza Dosa
- Kaju Paneer Cheese + Pizza Dosa
- Plain Dosa or Schezwan Butter Onion + Pizza Dosa

HIGH-LIFT “POWER PAIRINGS”

- Alleppey Special + Chinese Mushroom
- American Chopsuey + Sweet Corn Baby Corn
- Green Peas Chilli + Mix Veg Dosa

INSIGHTS

- Customers strongly prefer combination-based ordering.
- High-volume combos provide scalable business opportunities.

For example: Cheese Lovers Combo: Paneer Cheese + Mushroom Chilli Cheese, Gobi Cheese + Pizza Dosa, American Chopsuey Cheese + Pizza Dosa

- High-lift niche combinations can be used for limited-time offers.

BUSINESS RECOMMENDATION

- Create Pizza Dosa-based combo meals.
- Promote premium fusion bundles.
- Use high-lift niche pairings for targeted promotions.





Market Basket Analysis

1. HIGH-VOLUME ANCHOR PRODUCTS

The volume table shows that the strongest basket anchors are:

Product	Qty Sold With Basket Item
Pizza Dosa	62
Plain Dosa	42
Paneer Cheese	23
Mushroom Chilli Cheese	21
Paneer Chilli	15
Paneer Chinese Mushroom	15
Schezwan Butter Onion	15
Butter Plain Dosa	15

Insight
Pizza Dosa is the strongest anchor product, followed by Plain Dosa. These should be used as the base items for scalable combo meals because they already appear frequently in customer baskets.

2. STRONG SCALABLE COMBINATIONS

These are not necessarily the highest lift pairs, but they are more commercially useful because they appear across multiple bills.

Combo Pair	Bills With Both	Lift	Interpretation
Paneer Chilli + Pizza Dosa	6	2.25	Strong high-volume combo
Mushroom Chilli Cheese + Pizza	6	1.63	Popular fusion pairing
Mexican Salsa + Pizza Dosa	5	3.49	Strong cross-sell opportunity
Kaju Paneer Cheese + Pizza	5	3.05	Premium combo opportunity
Dubai Dosa + Paneer Cheese	4	3.4	Good fusion-pairing signal
American Chopsuey Cheese	4	2.79	Strong Indo-fusion pairing
Gobi Cheese + Pizza Dosa	4	2.79	Vegetarian combo opportunity
Mushroom Chilli Cheese + Paneer	4	2.45	Cheese-heavy combo pattern

Insight
For practical menu design, these are better combo candidates than extremely rare 254-lift pairs because they combine meaningful volume + positive association.

3. HIGH-LIFT NICHE “POWER PAIRINGS”

These pairs have very high Lift, meaning they are strongly associated when they appear together.

Combo Pair	Lift	Interpretation
Alleppey Special + Chinese Mushroom	254	Perfect exclusive pair
Aloo Plalak + Green Peas Dosa	127	Strong rare-pair association
American Chopsuey + Sweet Corn Baby	127	Niche bundle opportunity
Aloo Plalak + Schezwan Aloo Dry	84.67	Highly specific pairing
Green Peas Chilli + Mix Veg Dosa	63.5	Strong niche veg pairing
Green Peas Chilli + Masala Chilli	63.5	Strong spicy pairing
Chinese Mushroom + Schezwan Paneer	50.8	Strong Chinese-fusion pairing
Masala Cheese + Schezwan Baby	50.8	Strong spicy-cheese combo
Alleppey Special + Schezwan Paneer	50.8	Premium niche pairing

Important interpretation
These are not automatically the most profitable combos because many appear in only 1 bill. They are better described as: “exclusive niche pairings with high cross-sell strength, suitable for experimental or limited-time combo offers.”



Additional Business Insights & Limitations of Analysis

Additional Insights Discovered

- Fusion-style products outperform traditional products in revenue generation.
- Cheese-based and paneer-based products consistently appear in high-performing combinations.
- Evening hours generate the highest customer traffic and sales.
- Customers demonstrate strong preference for combination ordering behavior.
- Menu engineering opportunities exist through strategic combo design and upselling.

Limitations

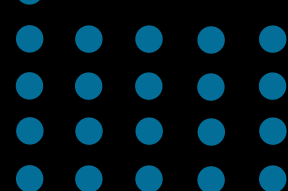
- The dataset covered a limited time period.
- Seasonal and long-term customer trends could not be evaluated.
- Customer demographic information was unavailable.
- Profit margins per product were not provided.
- Certain high-lift combinations were based on low-frequency transactions.

Despite these limitations, the analysis successfully identified meaningful customer ordering patterns and business opportunities.





Recommendations & Next Steps



Next Steps

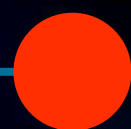
To make business analytics an integral part of decision-making, the restaurant should:

Recommendations

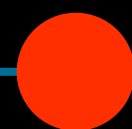
1. Introduce combo meals centered around Pizza Dosa and Plain Dosa.
2. Use high-performing fusion products in promotional campaigns.
3. Promote high-lift niche pairings as limited-time offers.
4. Optimize staffing and inventory during peak evening hours.
5. Use cross-selling recommendations in menus and billing systems.



Continuously collect and analyze transaction data.



Track customer purchase behavior regularly.



Monitor combo performance and profitability.



Build dashboards for real-time sales monitoring.



Use predictive analytics for demand forecasting and inventory planning.



Conclusion



This analysis successfully identified:

- Top-selling products and categories
- Customer ordering patterns
- Peak sales periods
- Strong cross-selling opportunities
- Profitable combo meal recommendations

The use of Power BI, DAX measures, and Market Basket Analysis helped transform raw restaurant sales data into actionable business insights.

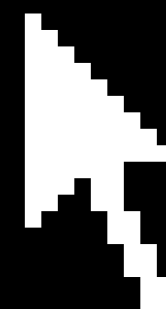
These findings can help the restaurant:

- Improve customer experience
- Increase average order value
- Optimize menu design
- Enhance operational decision-making
- Drive higher profitability through data-driven strategies





THANK YOU



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